Cerebro: Smart AI Companion Glasses

Project Description for Hackathons & Competitions

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1. Executive Summary

Cerebro is an open-source, AI-powered smart glasses project that provides hands-free intelligent assistance through voice commands, indoor navigation, and augmented reality overlays. Designed with privacy-first principles, Cerebro operates both online and offline, offering multi-professional support for engineers, students, medical professionals, and everyday users.

Tagline: "Your Intelligent Companion, Eyes Free"

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2. Problem Statement

2.1 The Challenge

- **Information Access:** People need quick access to information and assistance without stopping their activities
- **Accessibility:** Traditional devices require manual interaction, limiting with users disabilities or those with occupied hands
- **Navigation Complexity:** Indoor navigation in large buildings (hospitals, universities, airports) remains difficult without assistance
- **Privacy Concerns:** Cloud-based AI assistants continuously transmit personal data
- **Cost Barrier:** Existing smart glasses (Ray-Ban Stories, Google Glass Enterprise) are expensive and locked to ecosystems

2.2 Target Users

- Engineers and technicians needing hands-free technical support
- Medical professionals requiring quick reference during procedures
- Students navigating large campuses
- Visually impaired users needing spatial awareness
- Anyone seeking a lightweight, private AI companion

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3. Solution Overview

3.1 What Cerebro Does

Feature	Description
Voice Assistant	Natural language queries with STT and TTS
Indoor Navigation	QR code + graph-based pathfinding with AR arrows
Computer Vision	Scene understanding with Moondream vision model

Feature	Description
Smart Companion	Note-taking, reminders, task management
Privacy-First	Full offline operation capability
Modular Design	Hot-swappable components for easy maintenance

3.2 System Architecture

```
Smart Glasses (ESP32) --> Phone (Gateway) --> Cloud/Local LLM (Processing)

Components:
- Camera + Mic           - Voice + Vision Processing
- IMU Sensor             - AR Display
- Display                - WebSocket API
```

3.3 Key Features

Indoor Navigation System

- QR Code Detection using ArUco markers at known locations
- Graph Representation of buildings as nodes and edges
- Pathfinding using A* algorithm for optimal route calculation
- AR Guidance with directional arrows overlay on user's view

Voice Interaction Pipeline

```
User Speech --> WebRTC Audio --> Whisper STT --> LLM Agent --> Response
                                                                 |
                                                              TTS Audio
```

Computer Vision

- Scene understanding via Moondream vision model
- Object detection and description
- Face recognition (offline, privacy-preserving)

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4. Technical Implementation

4.1 Hardware Components

Component	Specification	Purpose
ESP32-S3	Dual-core, WiFi/BT	Main microcontroller

Component	Specification	Purpose
Camera	OV2640/OV5640	Computer vision input
IMU	MPU6050/ICM-20948	Orientation tracking
Microphone	I2S MEMS (INMP441)	Voice capture
Display	Micro-OLED/HUD	AR overlay output
Battery	LiPo 500mAh	Power supply

4.2 Software Stack

Layer	Technology
Frontend	Streamlit, Three.js
Backend	Flask, WebSocket
AI Agent	MCP Protocol, Moondream
Vision	OpenCV, ArUco Markers
Navigation	A* Algorithm, Graph Theory
STT/TTS	Whisper, Edge TTS
Hardware	Arduino/C++ (ESP32)

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5. Innovation & Novelty

5.1 What Makes Cerebro Unique

1. **Hybrid Operation Mode**
   - Online: Cloud AI for complex queries
   - Offline: Local models for privacy-critical tasks
2. **Modular Architecture**
   - Hot-swappable camera/mic modules
   - Easy component replacement
3. **Open-Source Ecosystem**
   - No vendor lock-in
   - Community-driven development
4. **Cost-Effective**
   - Target cost: <$100 (vs. $300+ competitors)
   - DIY-friendly design

5.2 Comparison with Competitors

Feature	Cerebro	Google Glass	Ray-Ban Stories
Price	~$100	$950+	$300+
Offline Mode	Yes	No	No
Open Source	Yes	No	No
Indoor Nav	Yes	No	No
Voice-First	Yes	Yes	Yes
AR Overlay	Yes	Yes	No

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6. Impact & Applications

6.1 Societal Impact

- **Accessibility:** Hands-free assistance for disabled users
- **Education:** Real-time information for students
- **Healthcare:** Reference support during procedures
- **Industrial:** Safety and maintenance guidance

6.2 Market Potential

- Global smart glasses market: $1.5B+ (2024)
- Growing demand for AI assistants
- Privacy-conscious consumer segment

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7. Project Deliverables

7.1 Hardware

- ESP32-S3 smart glasses prototype
- Custom PCB design (KiCad)
- 3D-printed enclosure

7.2 Software

- Voice assistant app (Streamlit)
- Navigation engine (Python)
- AR rendering pipeline (Three.js)
- Agent system (MCP-based)

7.3 Documentation

- Technical architecture diagram
- User manual
- Setup guide
- API documentation

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8. Competition Alignment

Why Judges Will Love This Project

Criterion	Cerebro Strength
Innovation	First open-source smart glasses with offline AI
Technical Depth	Full stack: hardware, firmware, software, AI
Practical Application	Real-world navigation and assistance
Completeness	End-to-end working system
Presentation	Live demo with voice and AR
Scalability	Modular, extensible architecture
Social Impact	Accessibility and privacy benefits

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9. Demo Script (3 Minutes)

1. **Opening (30s):** Show the glasses, explain the problem
2. **Voice Demo (45s):** "Take me to the library" --> Navigation starts
3. **AR Demo (45s):** Show AR arrows guiding the path
4. **Vision Demo (30s):** "What am I looking at?"
5. **Offline Demo (30s):** Switch to local mode
6. **Closing (15s):** Emphasize open-source and accessibility

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10. Tech Stack Summary

Languages: Python, C++, JavaScript, HTML/CSS **Frameworks:** Flask, Streamlit, Three.js, OpenCV **AI/ML:** Whisper, Moondream, LLM integration **Hardware:** ESP32, Arduino, KiCad **Communication:** WebSocket, HTTP, WebRTC

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11. Future Roadmap

- SLAM integration for markerless navigation
- Eye-tracking for hands-free control
- Multi-language support
- Wearable form factor refinement
- App ecosystem for third-party skills
- Enterprise deployment customization

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12. Abstract Summary (250 words)

Cerebro is an open-source smart glasses project that provides hands-free AI assistance through voice commands, indoor navigation, and AR overlays. The system addresses the growing need for accessible,

private, and affordable intelligent assistants.

Key features include:

- Voice-based interaction using Whisper STT and Edge TTS
- Indoor navigation using QR codes and A* pathfinding
- Computer vision with Moondream for scene understanding
- Privacy-first design with full offline operation capability
- Modular, open-source architecture with no vendor lock-in

The hardware consists of an ESP32-S3 microcontroller with camera, IMU, microphone, and display modules. The phone acts as a gateway for processing, connecting to both cloud and local AI models.

Cerebro targets engineers, students, medical professionals, and users with disabilities who need hands-free assistance. With a target cost under $100, it offers significant advantages over expensive competitors like Google Glass.

The project demonstrates expertise in embedded systems, computer vision, AI/ML, and full-stack development, making it ideal for hackathons and technical competitions.

Built with for accessibility, privacy, and innovation